

Empirical Proof Record: VALIDATED

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Proof ID: 80dd5c89019eb613e587a373f49395b5...

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1. Claim

Across 30 synthetic two-sample pairs of size 50 per group with varying effect sizes and variance ratios (seed=20260518), “`scipy.stats.ttest_ind(equal_var=False)`“, “`statsmodels.stats.weightstats.ttest_ind(usevar='unequal')`“, and “`pingouin.ttest(correction=True)`“ produce Welch t-statistics AND two-sided p-values that agree to $\leq 1e-09$ across all pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation; three-way variant with genuinely-independent statsmodels.)

- **Operationalisation:** Generate N (x, y) pairs where $x \sim \text{Normal}(0, \sigma_x^2)$ and $y \sim \text{Normal}(\delta, \sigma_y^2)$ for a sweep of δ across $[-1.0, 1.0]$ and variance-ratio σ_y/σ_x across $[0.5, 2.0]$. Compute Welch’s t and two-sided p under each backend. Compute $\text{max_abs_diff} = \max \text{ over } (\text{pair}, \text{statistic} \in \{t, p\}, \text{backend-pair}) \text{ of } |\text{stat_a} - \text{stat_b}|$. VALIDATED iff $\text{max_abs_diff} \leq \text{tolerance}$.
- **Threshold:** $\text{max_absolute_welch_difference} \leq 1e-09$ t_or_p
- **H0:** At least one backend pair disagrees on the Welch t-statistic OR the two-sided p-value by more than the floating-point tolerance.
- **H1:** All three backends compute identical Welch t AND p to floating-point tolerance across every pair.

2. Verdict

VALIDATED — observed $1.7763568394002505e-15$

30 pairs $\times$ 3 backends $\times$ 2 statistics; max pairwise $|\Delta| = 1.776e-15$ on (t, scipy_vs_statsmodels) ($\leq 1e-09$ tol)

3. Evidence

Pillar	Statistic	
welch_t_scipy_vs_statsmodels_vs_pingouin	max_absolute_welch_difference	1.776356839400250

4. Pre-registration

- Registered at: 2026-05-31T18:33:58.291000+00:00
- Config hash: 650f785bb6c4c650aa559236e9e29db3...
- Data hash: 650f785bb6c4c650aa559236e9e29db3...

5. Signature

95f8a8095cef7ae2e25188709a2eba2dce6b350701c83c6c9e6499a2c044986d